Delegate Background Guide

UNEP

UNITED NATIONS ENVIRONMENT PROGRAMME

Deliberating Upon the Rising Temperatures and Increasing Frequency of Extreme Heatwaves in Traditionally Cold Regions as a Consequence of Climate Change

BACKGROUND GUIDE

Prepared for delegates | MEA

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Letter from the Dais

Honourable Delegates,

Welcome to the United Nations Environment Programme (UNEP). This committee asks you to examine a climate challenge that often goes unnoticed: regions long associated with cold climates are now seeing rising temperatures, unusual heat events, and changes to ecosystems that their communities and infrastructure were never designed to withstand.

Your task is not simply to condemn climate change. We want you to investigate why these events are becoming more dangerous, how vulnerable communities can adapt, and how international cooperation can reduce future risks while respecting different national capacities. Think about the needs of Indigenous communities, local governments, scientists, workers, and countries whose economies depend on cold-region ecosystems.

We encourage you to debate with scientific care, practical imagination, and diplomatic respect. Strong proposals will be measurable, equitable, financially realistic, and attentive to both emissions reduction and adaptation.

Sincerely,
The Dais

Introduction to UNEP

UNEP is the United Nations' leading environmental authority. It supports international environmental cooperation, assesses the state of the environment, helps countries develop policy, and works with partners on climate change, biodiversity loss, pollution, and ecosystem degradation. UNEP does not replace national governments; it helps coordinate knowledge, partnerships, and action across borders.

On this agenda, UNEP's role is to connect climate science with practical environmental policy. Delegates can consider early-warning systems, resilient infrastructure, ecosystem protection, public-health preparedness, and climate finance. Keep two ideas distinct: **mitigation**, which means reducing greenhouse-gas emissions, and **adaptation**, which means preparing for impacts that are already occurring.

Introduction to the Agenda

Climate change is altering temperature patterns and raising the risk of extreme heat in many parts of the world. Traditionally cold regions, including high-latitude areas, mountain communities, and places with long winters, face particular risks because their buildings, roads, power systems, health services, and local livelihoods are often designed around historically cold conditions.

Heatwaves can increase heat-related illness and deaths, strain electricity and water systems, worsen wildfire conditions, accelerate permafrost thaw, and damage habitats. In cold regions, rapid warming can also affect snow and ice cover, river flows, transport routes, and the cultural practices of Indigenous peoples. The effects are not identical everywhere: geography, local infrastructure, public-health capacity, and the duration of heat events all matter.

> Avoid assuming that every unusual hot day is caused solely by climate change. Climate change alters the probability and intensity of many heat extremes, while individual events are assessed through scientific attribution methods. The committee should focus on risk reduction, reliable data, adaptation, and long-term emissions cuts.

Key Terms

TERM	DEFINITION
Heatwave	A period of unusually hot weather for a location, usually measured against local climate conditions over several days.
Climate mitigation	Actions that reduce greenhouse-gas emissions or increase their removal.

TERM	DEFINITION
Climate adaptation	Adjustments that reduce harm from current or expected climate impacts.
Permafrost	Ground that remains at or below 0°C for at least two consecutive years.
Early warning system	A system that detects hazards, communicates risk, and enables timely protective action.
Climate resilience	The ability of communities and systems to prepare for, withstand, recover from, and adapt to climate impacts.
Climate attribution	The scientific process of assessing how far human-caused climate change influenced the likelihood or intensity of a specific event.
Indigenous knowledge	Knowledge, practices, and relationships with the environment developed and maintained by Indigenous communities.

Key Issues

1. Public health and worker safety

Heat can cause dehydration, heat exhaustion, and heatstroke. Health systems in cold regions may have limited cooling capacity, and outdoor workers, older people, children, and people with existing illnesses face elevated risk.

2. Infrastructure built for cold conditions

Buildings may lack shading or cooling, and railways, roads, bridges, and power equipment can be affected by heat outside their historical design range. Adaptation must avoid creating excessive energy demand.

3. Permafrost, snow, and ice

Warming can destabilize frozen ground, damage foundations and pipelines, alter water systems, and affect habitats and travel routes. Monitoring and locally appropriate construction standards are important.

4. Wildfire and ecosystem change

Hotter and drier conditions can raise fire danger in some regions and shift species ranges. Fire policy must combine prevention, land management, community preparedness, and ecological knowledge.

5. Unequal capacity to adapt

Small towns, remote settlements, and Indigenous communities may face high costs and limited access to health care, technical expertise, and emergency services. Funding and decision-making should include the affected communities.

6. Mitigation and energy demand

Cooling can protect lives but may increase electricity demand and emissions if it is powered by fossil fuels. Passive cooling, efficient buildings, clean energy, and heat-action plans can reduce this trade-off.

Major Stakeholders

National governments

Set climate policy, public-health standards, infrastructure codes, and emergency plans.

UNEP and UN climate bodies

Support environmental cooperation, assessments, and the implementation of international commitments.

World Meteorological Organization (WMO)

Coordinates weather and climate services and supports early-warning systems.

World Health Organization (WHO)

Provides public-health guidance on heat risks and climate-related health impacts.

Local and regional authorities

Run emergency response, public cooling spaces, urban planning, and local infrastructure.

Indigenous peoples and local communities

Contribute local knowledge and identify culturally appropriate adaptation priorities.

Scientists and universities

Provide climate monitoring, attribution research, modelling, and risk assessment.

Development banks and climate funds

Can help finance adaptation and resilient infrastructure.

Past International Actions and Frameworks

UN Framework Convention on Climate Change (1992)

Established the main international framework for climate cooperation.

Kyoto Protocol (1997)

Set binding emissions targets for participating developed countries under its first commitment period.

Paris Agreement (2015)

Sets the global framework for limiting warming and strengthening adaptation and climate resilience.

Sendai Framework for Disaster Risk Reduction (2015–2030)

Promotes understanding risk, strengthening governance, investing in resilience, and improving preparedness.

Early Warnings for All initiative (launched 2022)

Seeks universal protection through effective early-warning systems.

UNEP Emissions Gap and Adaptation Gap reports
Assess the gap between current action and the action needed to address climate change and adaptation needs.

Timeline

DATE	DEVELOPMENT
1992	UNFCCC adopted at the Rio Earth Summit.
1997	Kyoto Protocol adopted.
2015	Paris Agreement and Sendai Framework adopted.
2018	IPCC special report examines impacts associated with 1.5°C of global warming.
2021	IPCC Sixth Assessment Report highlights widespread human influence on the climate system.
2022	Early Warnings for All initiative launched.
2023	IPCC completes its Sixth Assessment cycle, emphasizing rapid, deep emissions reductions and adaptation.
2024 onward	Countries continue to update climate plans and strengthen heat preparedness; delegates should verify the latest national data before citing specific events.

Key Questions a Resolution Must Answer

1. How can countries improve heatwave forecasting and ensure warnings reach remote communities?
2. What minimum protections should exist for workers and vulnerable people during extreme heat?
3. How can cold-region infrastructure be retrofitted without locking communities into high-emission cooling?
4. How should international finance support countries and communities with limited adaptation capacity?
5. How can Indigenous knowledge be included with free, prior, and informed participation?
6. What data should countries share to compare heat risks and evaluate whether policies work?

Bibliography and Further Reading

Use these official and institutional sources as starting points, and check the latest documents before citing statistics or claiming that a policy is current.

UN Environment Programme – Climate Change
https://www.unep.org/explore-topics/climate-action

World Meteorological Organization – Climate
https://wmo.int/topics/climate

IPCC Sixth Assessment Report
https://www.ipcc.ch/assessment-report/ar6/

UNFCCC – The Paris Agreement
https://unfccc.int/process-and-meetings/the-paris-agreement

WHO – Climate Change and Health
https://www.who.int/news-room/fact-sheets/detail/climate-change-and-health

UNDRR – Sendai Framework
https://www.undrr.org/implementing-sendai-framework/what-sendai-framework

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